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Method and system for automatic detection and recognition of a digital image

Abstract

An automatic measuring system containing configurable integrated circuits is able to process information via captured images. The automatic measuring system includes a metering instrument, a camera, a recognition module, and a localization module. The metering instrument has at least one display for visually displaying a number and measures the amount of measurable substance or resources (i.e., electricity and water) consumed. The camera captures an image of the number representing at least a portion the amount of measurable substance. The recognition module is operable to generate a value in response to the image and the coordinates wherein the coordinates are used to decode the image via restoring captured image to the original readout counter value. The localization module is removably or remotely coupled to the camera and operable to generate the coordinates in accordance with the image captured by the camera.

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Background/Summary

PRIORITY (1) This patent application is a continuation patent application of a U.S. patent application having a U.S. patent application Ser. No. 18/126,263, filed on Mar. 24, 2023 in the name of same inventors and entitled “Method and System for Automatic Detection and Recognition of A Digital Image,” issued into a U.S. patent with a U.S. Pat. No. 11,967,062, on Apr. 23, 2024, which is also a continuation patent application of a U.S. patent application having a U.S. patent application Ser. No. 17/489,749, filed on Sep. 29, 2021 in the name of same inventors and entitled “Method and System for Automatic Detection and Recognition of A Digital Image,” issued into a U.S. patent with a U.S. Pat. No. 11,615,522, on Mar. 28, 2023. All mentioned U.S. applications are hereby incorporated by reference.

FIELD

(1) The exemplary embodiment(s) of the present invention relates to the field of programmable semiconductor devices for computer hardware and software. More specifically, the exemplary embodiment(s) of the present invention relates to a system containing a field-programmable gate array (“FPGA”) capable of automatically reading a meter via a captured image.

BACKGROUND

(2) A traditional electro-mechanical meter, such as an electricity meter, electric meter, electrical meter, energy meter, kilowatt-hour meter, gas meter, water meter, and/or pressure meter, is a measuring instrument or device that measures for example the amount of electric energy or water consumed by a residence, a business, or an electrically powered apparatus. For example, utility companies use electric meters installed at customers' premises for purposes of billing and/or monitoring. A drawback relating to a traditional or conventional electro-mechanical meter is that the output values or counted values of the meter need to be read out manually. For example, taking the meter reading manually on a regular basis is common, but the task of manual reading a meter can be tedious, costly, and prone to errors.

(3) A conventional approach to overcome the manual reading is the creation of smart meter which, for example, can generate and transmit pulses to meter interfacing unit so that the counter can be accurately and instantaneously acquired. A problem associated with using smart meters is that the replacing hundreds and/or thousands of existing traditional meters can take a lot of time and effort.

SUMMARY

(4) On embodiment of the presently claimed invention discloses an automatic measuring system (“AMS”) using one or more configurable integrated circuits (“ICs”) for processing information via captured images. AMS includes a metering instrument, a camera, a recognition module, and a localization module. The metering instrument, having at least one display for visually displaying a number, is able to measure the amount of consumable and/or substance consumed by a customer within a period of time. For example, consumable or substance can be electricity, gas, and/or water. The camera, having a lens, is used to capture an image containing one or multiple numbers indicating that the amount of consumable and/or substance that has been consumed. The recognition module is used to calculate a counter value indicating the amount of substance that has been consumed based on the image and coordinates. The coordinates are used to facilitate restoring the readout counter image captured in the image to the original readout counter value displayed by the meter. The localization module which can be removed from AMS. A function of localization module is to generate a set of coordinates according to the image captured by the camera.

(5) Additional features and benefits of the exemplary embodiment(s) of the present invention will become apparent from the detailed description, figures and claims set forth below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The exemplary embodiment(s) of the present invention will be understood more fully from the detailed description given below and from the accompanying drawings of various embodiments of the invention, which, however, should not be taken to limit the invention to the specific embodiments, but are for explanation and understanding only.

(2) FIG. 1 is a block diagram illustrating an AMS employing one or more FPGA and/or wireless accessible PLD (“WAP”) to perform one or more automatic meter readings in accordance with one embodiment of the present invention;

(3) FIGS. 2A-2B are block diagrams illustrating an exemplary meter displaying a sequence of numbers observed by a camera in accordance with one embodiment of the present invention;

(4) FIG. 3 is a block diagram illustrating a localization module or DLM and a camera for generating coordinates associated to a captured image in accordance with one embodiment of the present invention;

(5) FIG. 4 is a block diagram illustrating a recognition module or DRM configured to generate the counter value in accordance with one embodiment of the present invention;

(6) FIG. 5 is a block diagrams illustrating measuring system or AMS using a two-module approach

in accordance with one embodiment of the present invention;

(7) FIGS. **6-7** are block diagrams illustrating a wireless accessible PLD (“WAP”) capable of providing wireless FPGA for hosting at least a portion of the measuring system in accordance with one embodiment of the present invention in accordance with one embodiment of the present invention;

(8) FIGS. **8-9** are diagrams illustrating an FPGA containing one or more WCBs capable of being programmed to perform at least some functions of a measuring system in accordance with one embodiment of the present invention;

(9) FIG. **10** is a flowchart illustrating a process of obtaining a counter value via a measuring system in accordance with one embodiment of the present invention; and

(10) FIGS. **11-12** are diagrams illustrating a digital processing system and a cloud-based system environment using one or more measuring systems having FPGAs in accordance with one embodiment of the present invention.

DETAILED DESCRIPTION

(11) Embodiments of the present invention disclose a method(s) and/or apparatus for providing an automatic measuring system (“AMS”) capable of reading one or more measuring meters.

(12) The purpose of the following detailed description is to provide an understanding of one or more embodiments of the present invention. Those of ordinary skills in the art will realize that the following detailed description is illustrative only and is not intended to be in any way limiting. Other embodiments will readily suggest themselves to such skilled persons having the benefit of this disclosure and/or description.

(13) In the interest of clarity, not all of the routine features of the implementations described herein are shown and described. It will, of course, be understood that in the development of any such actual implementation, numerous implementation-specific decisions may be made in order to achieve the developer's specific goals, such as compliance with application- and business-related constraints, and that these specific goals will vary from one implementation to another and from one developer to another. Moreover, it will be understood that such a development effort might be complex and time-consuming but would nevertheless be a routine undertaking of engineering for those of ordinary skills in the art having the benefit of embodiment(s) of this disclosure.

(14) Various embodiments of the present invention illustrated in the drawings may not be drawn to scale. Rather, the dimensions of the various features may be expanded or reduced for clarity. In addition, some of the drawings may be simplified for clarity. Thus, the drawings may not depict all of the components of a given apparatus (e.g., device) or method. The same reference indicators will be used throughout the drawings and the following detailed description to refer to the same or like parts.

(15) In accordance with the embodiment(s) of present invention, the components, process steps, and/or data structures described herein may be implemented using various types of operating systems, computing platforms, computer programs, and/or general-purpose machines. In addition, those of ordinary skills in the art will recognize that devices of a less general-purpose nature, such as hardware devices, field programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), or the like, may also be used without departing from the scope and spirit of the inventive concepts disclosed herein. Where a method comprising a series of process steps is implemented by a computer or a machine and those process steps can be stored as a series of instructions readable by the machine, they may be stored on a tangible medium such as a computer memory device (e.g., ROM (Read Only Memory), PROM (Programmable Read Only Memory), EEPROM (Electrically Erasable Programmable Read Only Memory), FLASH Memory, Jump Drive, and the like), magnetic storage medium (e.g., tape, magnetic disk drive, and the like), optical storage medium (e.g., CD-ROM, DVD-ROM, paper card and paper tape, and the like) and other known types of program memory.

(16) The term “system” or “device” is used generically herein to describe any number of

components, elements, sub-systems, devices, packet switch elements, packet switches, access switches, routers, networks, computer and/or communication devices or mechanisms, or combinations of components thereof. The term “computer” includes a processor, memory, and buses capable of executing instruction wherein the computer refers to one or a cluster of computers, personal computers, workstations, mainframes, or combinations of computers thereof.

(17) On embodiment discloses an AMS containing one or more configurable ICs or FPGAs capable of processing information based on captured images. AMS includes a meter, a camera, a recognition module, and a localization module. The meter, having at least one display for visually displaying a number, measures the amount of consumable and/or measurable substance. For example, consumable or measurable substance can be electricity, gas, steam, heat, thermal liquid, alcohol, and/or water. The camera is used to capture one or more images wherein each image may contain one or more numbers indicating the amount of substance or resource consumed. The recognition module calculates a counter value indicating the amount of substance consumed based on the image and coordinates. The coordinates, in one embodiment, are used to facilitate decoding or restoring the readout counter image shown in the image to an original readout counter value displayed by the meter. The localization module provides the coordinates for calibration and it is removed from AMS after the calibration.

(18) FIG. 1 is a block diagram illustrating an AMS **100** employing one or more FPGA and/or wireless accessible PLD (“WAP”) to perform one or more automatic meter readings in accordance with one embodiment of the present invention. AMS **100** includes a meter **102**, a camera **106**, a localization module **110**, and a recognition module **112**. In one example, AMS **100** further includes a rigid camera mount **104** used to anchor camera **106** to a fixed location allowing camera **104** has a field of view **118** of display **116**. It should be noted that the underlying concept of the exemplary embodiment(s) of the present invention would not change if one or more blocks (circuit or elements) were added to or removed from FIG. 1.

(19) Meter **102**, also known as a metering instrument or measuring instrument, is used to count or measure the amount of measurable substance or resources consumed by a residential home, business entity, and/or electronic device. The measurable substance, moving substance, flowing substance, consumable resources, and/or substance can be, but not limited to, electricity, natural gas, water, chemical solutions, air flow, air pressure, thermal liquid, and the like. Meter **102** includes a display **116** which can be structured in various configurations, such as a sequence of digits, a group of dials, a serial numbers, a screen displaying a value, a bar, a line, and so on. The digits or numbers displayed on display **116** are the original readout or readout counter value. For example, display **116** displays or projects **10,000** which indicates a readout counter value of 10 kilowatt of electricity that has been consumed.

(20) Meter **102**, in one example, can be an electro-mechanical meter, electricity meter, electric meter, electrical meter, energy meter, kilowatt-hour meter, gas meter, displacement water meter, pressure meter, or the like. A function of meter **102** is to measure or count the amount of resource or substance such as electricity consumed by a customer or device. Another function of meter **102** is to assist a supplier such as a power company to manage the operation. For example, a municipal water company installs electric meters at customers' premises for billing as well as supply management.

(21) Camera **106** is an optical instrument capable of capturing one or more images or videos via its lens through a field of view **118**. For example, camera **106**, which includes a lens and aperture, captures an image or a serial of images based on orientation of its lens that has a field of view **118**. Camera **106**, in one aspect, is a semiconductor imager or optical sensor capable of being fabricated via a semiconductor process. Image sensors can be fabricated by chips and/or together as part of FPGA die. Camera **106**, in one embodiment, is placed, fixed, or anchored at a predefined distance from display **116** of meter **102** whereby a field of view **118** between display **116** and camera **106** is established. Field of view **118** illustrates an exemplary distance between display **116** and camera

106 in which the lens of camera **106** capturing its view of surrounding environment. When field of view **118** includes display **116** of meter **102**, camera **106** can capture an image of readout counter value displayed by display **116**.

(22) Localization module **110**, also known as digit localization module (“DLM”) or localization unit, is structured as a removable unit or module capable of being detached or separated from AMS (or camera **106**) after the coordinate(s) or digit location is calculated or calibrated. For example, the coordinates, also known as location coordinates, digit locations, and/or counter orientation, is a set of physical coordinates and angle(s) which are used to compensate image distortion due to meter installation and/or meter types. To identifying the coordinates, DLM **110**, in one embodiment, employs a multi-task model and a processor to calculate digit coordinates which can be used to correct the distorted digits in the captured image.

(23) DLM **110** further includes a camera interface for facilitating receipt of images from a camera-to-localization (“CL”) bus **122**. DLM **110** also includes a recognition interface for transmitting the coordinates to DRM **112** via a localization-to-recognition (“LR”) bus **126**. Depending on the applications, CL bus **122** and LR bus **126** can either be cables, wires, wireless links, and/or a combination of cables, wires, and wireless links. After calibration, both CL bus **122** and LR bus **126** can be disconnected or removed whereby DLM **110** can be separated from AMS **100**. An FPGA, in one aspect, is used to host DLM **110**.

(24) Recognition module **112**, also known as digit recognition module (“DRM”), is capable of generating counter value **130** via an output **128** based on images received from camera **106** and the coordinates received from DLM **110**. In one aspect, upon receipt of the coordinates (or digit location) from DLM **110** via LR bus **126**, DRM **112** stores the coordinates in a local storage or memory. Since field of view **118** is fixed by rigid camera mount **104**, the coordinates can be used repeatedly whereby additional coordinates calculation and/or calibration is not required.

(25) DRM **112** includes an interface for facilitating receipt of images from a camera-to-recognition (“CR”) bus **120**. DRM **112** also includes a recognition interface for receiving the coordinates from DLM **110** via LR bus **126**. Depending on the applications, CR bus **120** and LR bus **126** can either be cables, wires, wireless links, and/or a combination of cables, wires, and wireless links. In one aspect, camera **106** and recognition module **112** can be fabricated or structured or placed in a single device, IC, or semiconductor die. In one embodiment, DRM **112** can be embedded in an FPGA.

(26) In operation, camera **106**, based on field of view **118**, captures an image containing multiple digits representing readout counter value generated by meter **102**. The image or images are forward to DLM **110** via CL bus **122** for coordinate calculation or calibration. The image or images are also forward to DRM **112** via CR bus **120** for output counter output **130**. Upon receipt of the coordinates from DLM **110**, DRM **112** generates counter value **130** which should be the same or substantially the same to the readout counter value displayed by display **116**.

(27) AMS **100**, in one embodiment, is configured to use its communication block to periodically communicate with a predefined entity such as a power company or a water company for collecting and/or managing commodity and/or resource supply. For example, a power company can use AMS **100** to record the power usage of each customer over a period of time. The communication block of AMS **100** can be land (wire) based, wireless based, or a combination of land and wireless based communication network. For example, a wireless communication block within an FPGA can be configured to provide network communications for AMS **100**.

(28) Alternatively, meter **102**, which can be an electromechanical meter, includes a display **116** configured to display its reading (i.e., counter) or readout counter value indicating the amount of substances such as electricity or water that has been consumed. Camera **106** is attached to meter **102** through rigid camera mount **104** so that field of view **118** of camera **106** can cover the entire counter region of meter **102**. Captured image is transferred to both DLM **110** and DRM **112**. It should be noted that the position, size, orientation (i.e., angle of rotation) of counter in the captured image is difficult to control partially because the distorted counter in the captured image may

depend on the installation of each individual meter. Each distorted counter in the captured image may vary between the meters. Accordingly, DLM **110** is used to detect the counter and locate the digits during the calibration process during the setup process. Once completed, DLM **110** can be removed. Unless the position of camera **106** relative to the meter changes frequently, permanent installation of DLM **110** to AMS **100** is not necessary. DRM **112**, which receives and saves the digit location from DLM **110**, recognizes each digit and outputs the counter value **130**.

(29) In another exemplarily embodiment, AMS or system **100**, containing configurable ICs or FPGAs for processing information via captured images, includes a metering instrument, a camera, a recognition module. The metering instrument such as meter **102** containing at least one display for visually displaying a number as a readout counter value is configured to record amount of measurable substance within a predefined time period. The measurable substance is electricity, natural gas, oil, air, heat, pressure, steam, and water. The metering instrument can be an electrical meter for metering usage of electricity, gas meter for measuring usage of natural gas, thermal meter for metering heat or cold, steam meter for measuring steam, or water meter for counting usage of water.

(30) The camera such as camera **106** is used to capture an image containing number(s) representing at least a portion of measurable substance. The camera is an optical device having a lens capable of capturing one or more images displayed by the display of the metering instrument. In one application, the camera is configured to capture a sequence of digits displayed by the display.

(31) The recognition module such as DRM **112** is operable to generate a counter value in response to the image and coordinates which is used to minimize image distortion. AMS **100** further includes a localization module such as DLM **110** which is removably coupled to the camera and/or AMS **100**. A function of localization module is to generate the coordinates in accordance with the image captured by camera **106**.

(32) The localization module such as DLM **110** further includes an interface, multi-task model, and controller (or CPU) wherein the interface is used to facilitate removably coupling to the camera via a communication channel such as CL bus **122** for receiving the image from camera **106**. The multi-task model block is operable to detect at least one digit from the image and an angle prediction. The controller, in one aspect, is configured to generate a set of coordinates at least partially based the angle prediction. Note that the communication channel such as CL bus **122** can be a wireless connection via a local onboard wireless transceiver block.

(33) The localization module further includes a recognition interface used to facilitate removably coupling or connection to the recognition module via a communication channel such as LR bus **126** for supplying the coordinates to the recognition module. An FPGA may be programmed to host the localization module. It should be noted that LR bus **126** can be a wireless connection via a local onboard wireless transceiver block. In one aspect, the localization module such as DLM **110** further includes an image cropping block and an image rotation block. The image cropping block is used to crop boundaries of the image based on the coordinates and the image rotation block is capable of rotating the image based on the angle prediction.

(34) In one embodiment, an FPGA is used to perform functions provided by the recognition module such as DRM **112**. The recognition module includes a buffer and an interface wherein the buffer is used to temporarily store one or more images representing one or more digital numbers from the camera. The interface is used to manage communication with a localization module via RL bus **126** for receiving information relating to the coordinates. The coordinates include information relating to digit locations and angle of rotation. The recognition module includes a local memory for storing the coordinates. In one aspect, the recognition module includes an image cropping block and an image rotation block. The image cropping block is operable to crop boundaries of the image based on at least one digit location. The image rotation block is operable to rotate the image based on the angle of rotation.

(35) An advantage of using AMS **100** is that localization module **110** can be removed from AMS

after image calibration whereby simplify overall AMS configuration.

(36) FIG. 2A is a block diagram **200** illustrating an exemplary meter displaying a sequence of numbers observed by a camera in accordance with one embodiment of the present invention. Diagram **200** includes a meter **202** and a camera **206**. Camera **206** includes a lens **220** which is oriented in such a way that its field of view **118** is focused on display **216** of meter **202**. It should be noted that the underlying concept of the exemplary embodiment(s) of the present invention would not change if one or more blocks (circuit or elements) were added to or removed from diagram **200**.

(37) Meter **200**, for example, includes display **216** capable of displaying a sequence of eight (8) digits or numbers representing a readout counter value of a water meter as indicated by numeral **210**. While the largest digit represents 10,000 kiloliter of water, the smallest digit represents one (1) liter of water. The eight (8) digits or number can either be mechanical numbers or electronic digital display. In another example, meter **200** includes display **216** capable of displaying a group dials representing a reading of an electrical meter as indicated by numeral **212**. While the largest dial represents 10,000 kilowatt hour ("kWh") of electricity, the smallest dial represents 0.1 kWh. The dials can either be mechanical dials or electronic images as dial configuration.

(38) It should be noted that meter display **216** can be arranged in various other illustrations, such as, but not limited to, bar scales, pie scales, letters, waves, and the like. Meter **202**, which can be an electro-mechanical meter, is used as an electricity meter, electric meter, electrical meter, energy meter, kilowatt-hour meter, gas meter, water meter, and/or pressure meter for measuring the amount of substances or resources consumed by an entity or device.

(39) FIG. 2B illustrates block diagrams **250-252** showing discrepancy or distorted images from non-distorted reading number or readout counter value displayed by the meter in accordance with one embodiment of the present invention. Diagram **250** illustrates a captured image **260** and a restored image **262** after cropping and rotating. Similarly, diagram **252** illustrates a distorted image **270** captured by the camera and a restored image **272** after cropping and rotating. It should be noted that a readout counter image **256** is a digitized electronic copy of a readout counter value displayed by the meter. Depending on the processing of picture taking as well as field of view, the readout counter image **256** may be a distorted version of the readout counter value. For example, the readout counter image **256** could be rotated certain angles such as θ° rotated from the original readout counter value displayed by the meter.

(40) Upon capturing or photographing an image by a camera with its lens aim at a field of view focusing at the display of meter, an image or picture **260** is obtained. While image **260** presents a distorted meter reading, image **260** is forwarded to DLM and DRM via wired or wireless transmissions. After receiving image **260** from the camera (not shown in diagram **250**), DLM identifies a bounding box **266** with x, y, w, h and angle θ (theta) **268** for setting a minimal boundary for a distorted readout counter image **256**. It should be noted that a bounding box is an imaginary rectangle box that is used for a reference point of an object. Angle θ **268** is an angular projection indicating a degree of readout counter image **256** captured in image should be rotated to restore to the original readout counter value displayed by the meter.

(41) A function of DLM, in one aspect, is to identify a set of coordinates which can be used to restore readout counter image **256** to the original or substantially similar to the original readout counter value as indicated by numeral **258**. The coordinates include, but not limited to, $x_i, y_i, w_i, h_i, \theta_i$ for a bounding box of each digit, number, or dial. To accurately obtain or identify a readout counter value, the coordinates are needed to facilitate cropping, rotating, and scaling to restore readout counter image **256** to its non-distorted or less distorted counter value. For example, upon detecting bounding box **266**, DLM generates $x_1', y_1', w_1', h_1', \theta_1$ for first digit "1" as indicated by numeral **268**. Once the bounding box of each digit (i.e., x_i, y_i, w_i, h_i) is identified, the coordinates (i.e., x_i, y_i, w_i, h_i) are mapped back to the captured image domain (i.e., x_i', y_i', w_i', h_i') for re-orientated or restored the readout counter image to its original orientation.

(42) The distortion of readout counter value could be caused by various reasons. For example, installation of the meter, type of meter, model of measuring instrument, aging of the meter, and noise during image transmission. To restore the readout counter image, DLM forwards the coordinates to DRM as indicated by numeral **266**. It should be noted that transmitting information of coordinates from DLM to DRM can be wired or wireless network transmission.

(43) After receiving the coordinates from DLM and captured image **260** from the camera, DRM, in one embodiment, crops, rotates, and scales each digit in image **260** in response to the coordinates. After cropping, rotating, and scaling, DRM generates a restored image **262** containing re-orientated readout counter value **258** with updated coordinates $x1, y1, w1$, and $h1$. The counter value is recognized or obtained after the restored image **262** is accomplished.

(44) An advantage of using DLM and DRM is that it is relatively simple and cost effective method to automatically read a meter without the meter reader manually reads the meter.

(45) Diagram **252** is similar to diagram **250** except that instead of numeral or digital representation, dials are used for the meter. Upon capturing an image **270** by a camera, the image **270** containing readout counter image using dials is transmitted to DLM and DRM. After identifying the coordinates associated with the dials within the bounding box **278**, the coordinates (i.e., $xi', yi', wi', hi', \theta$) are forwarded from DLM to DRM as indicated by numeral **276**. Upon restoring or re-orienting readout counter image, the counter value is identified based on the restored readout counter image **272**.

(46) FIG. **3** is a block diagram **300** illustrating a localization module or DLM and a camera for generating coordinates associated to a captured image in accordance with one embodiment of the present invention. Camera **106** and DLM **110** are connected via CL bus **122** wherein camera **106** is responsible to take pictures or images in accordance with its field of view. A function of DLM **110** is to generate a set of coordinates for calibrating AMS based on images from camera **106**. It should be noted that the underlying concept of the exemplary embodiment(s) of the present invention would not change if one or more blocks (ICs or elements) were added to or removed from diagram **300**.

(47) DLM **110**, in one embodiment, includes a frame buffer **302**, an image cropping unit **304**, an image rotation unit **306**, an image scaler **308**, a digit detection model **310**, a central processing unit (“CPU”) **316**, and a multi-task model **312**. Frame buffer **302**, in one example, can be a semiconductor memory device capable of electronically storing digital information. The semiconductor memory includes, but not limited to, random-access memory (“RAM”), static RAM (“SRAM”), dynamic RAM (“DRAM”), flash memory, phase-change memory, magnetoresistive RAM (“MRAM”), and the like. Upon receipt of captured images from camera **106**, DLM stores the captured images in frame buffer **302**. The captured images, which include the readout counter images, are subsequently forwarded to multi-task model **312** and image cropping unit **304**. It should be noted that a readout counter image is a digitized electronic copy of a readout counter value displayed by the meter. Depending on the processing picture taking as well as field of view, the readout counter image may be a distorted version of the readout counter value.

(48) Multi-task model **312**, in one example, is a Machine Learning (“ML”) model which can be used to optimize a particular metric or task via learning and repetition process. To optimize a task, a single ML model or a cluster of models are used for training output a desirable outcome. To improve task performance such as identifying original orientation of readout counter value, multi-task model **312**, in one aspect, shares representations of related tasks to improve performance. A function of multi-task model **312** is to detect the counter within the image and identify whether the counter in the image is distorted or not. If the distortion is detected, multi-task model **312** predicts a likelihood angle of rotation needed to restore the counter (readout counter image) to its original readout counter value displayed by the meter.

(49) Multi-task model **312**, in one example, includes a multi-task learning model used to facilitate for object detection (for position and size) such as YOLO (You Only Look Once) model or SSD

(Single Shot MultiBox Detector) model as well as classification or regression (for orientation) purposes. After calculating and/or identifying the angle of rotation and/or other attributes, the angle such as θ and attributes are forwarded to CPU **316**. The attributes (i.e., xi' , yi' , wi' , hi' , θ) indicate the discrepancy or distortion between the original readout counter value by the meter and the counter in the image.

(50) Image cropping unit **304**, in one embodiment, is used to identify a counter location in view of the captured image. The counter location, for example, is a minimal boundary around the readout counter image. Upon receipt of the captured image, image cropping unit **304** identifies the bounding box containing distorted readout counter value based on the counter location and angle of rotation received from multi-task model **312** via CUP **316**. After cropping, the cropped image containing the bounding box is forwarded to image rotation unit **306**.

(51) Image rotation unit **306** is configured to rotate the distorted readout counter value or readout counter image within the bounding box to a non-distorted readout counter value or original readout counter value displayed by the meter based on the angle of rotation. The angle of rotation and/or attributes are from CPU **316**. Note that the angle and coordinates are obtained from multi-task model **312** as indicated by numeral **330**. The cropped and rotated readout counter image is forwarded to image scaler **308**.

(52) Image scaler **308** is used to resize or scale the cropped and rotated readout counter image to a predefined size and/or scale. After scaling, the scaled readout counter image is forwarded to digit detection model **310**. Note that the scaled readout counter image should be the same or substantially the same as the readout counter value.

(53) Digit detection model **310**, in one aspect, is used to detect the digit, digits, number, numbers, dial, dials, and the like in view of the scaled readout counter image. Based on detected digit or digits, digit detection model **310** generates a set of coordinates **328** in accordance with the attributes and angle from CPU **316** as indicated by numeral **332**.

(54) CPU **316**, also known as central processor, main processor, processor, or controller, is capable of performing a set of computational functions based on a set of executable instructions. CPU **316**, in one embodiment, includes a neural processing unit (“NPU”) to assist multi-task model **312** for performing computational tasks more efficiently. Also, CPU **316** is used to manage the data flows within DLM **110**.

(55) DLM **110**, in one example, also includes communication interfaces **320-322**. Interface **320** is used to facilitate data transmission such as captured images between camera **106** and DLM **110** via communication channel or CL bus **122**. In one example, interface **320** is capable of facilitating data transmission via wires or cables. Alternatively, interface **320** is configured to facilitate data transmission via wireless communication. Interface **322** is used to facilitate data transmission such as coordinates **328** between DLM **110** and DRM via communication channel or LR bus **326**. In one example, interface **322** is capable of facilitating data transmission via wires or cables. Alternatively, interface **322** is configured to facilitate data transmission via wireless communication.

(56) DLM **110**, in one embodiment, is capable of being removed from AMS after the calibration process. Since the distance (or field of view) between camera **106** and the display of meter is fixed, the coordinates generally do not change. Accordingly, DLM **110** is not needed for AMS once the calibration process (or coordinates) is done.

(57) An FPGA, in one embodiment, is used to host DLM **110**. For example, FPGA can be programmed to embed DLM **110**. Since DLM **110** contains multi-task model **312**, a relatively large size of FPGA may be required to handle multi-task machine learning. If wireless communications are used, a wireless FPGA or WAP can be used to host DLM **110**.

(58) An advantage of using wireless communications for CL bus **122** and LR bus **326** is that it is relatively easy for connecting or disconnecting DLM to or from AMS.

(59) FIG. **4** is a block diagram **400** illustrating a recognition module or DRM configured to generate the counter value **130** in accordance with one embodiment of the present invention.

Diagram **400** includes camera **106** and DRM **112** wherein camera **106** and DRM **112** are connected via CR bus **120**. Camera **106** is responsible to take pictures or capture images in accordance with its field of view. A function of DRM **112** is to generate a sequence of digits as counter value **130** for AMS based on images from camera **106**. It should be noted that the underlying concept of the exemplary embodiment(s) of the present invention would not change if one or more blocks (ICs or elements) were added to or removed from diagram **400**.

(60) DRM **112**, in one embodiment, includes a frame buffer **402**, image cropping unit **406**, image rotation unit **408**, image scaler **410**, digit classification model **412**, and CPU **416**. Frame buffer **402**, in one example, is a semiconductor memory device capable of electronically storing digital information. The semiconductor memory includes, but not limited to, random-access memory (“RAM”), static RAM (“SRAM”), dynamic RAM (“DRAM”), flash memory, phase-change memory, magnetoresistive RAM (“MRAM”), and the like. Upon receipt of the captured images from camera **106**, DRM **112** stores the captured images in frame buffer **402**.

(61) DRM **112**, in one aspect, includes communication interfaces **420-422**. Interface **420** is used to facilitate data transmission such as the coordinates (i.e., $x1'$, $y1'$, $w1'$, $h1'$ and θ) of one or more digits from DLM via communication channel or LR bus **126**. Interface **420** is capable of facilitating data transmission via wires or cables. Alternatively, interface **420** can be configured to facilitate data transmission via wireless communication. Interface **422** is used to facilitate data transmission output such as counter value via a communication channel or an output channel **432**. In one example, interface **422** outputs counter value **130** to user(s) via wires or cables. Alternatively, interface **422** transmits counter value **130** via wireless communication network. Upon receipt of the coordinates from DLM, the coordinates are subsequently passed to CPU **416**.

(62) CPU **416**, which can be a central processor, main processor, processor, or controller, is capable of performing a set of functions based on executable instructions. CPU **416**, in one embodiment, includes an NPU for assisting digit classification model **412** for enhancing overall performance. One function of CPU **416** is to distribute the information of coordinates to various blocks such as image cropping unit **406** and/or image rotation unit **408**. Another function of CPU **416** is to manage data flows within DRM **112**.

(63) Image cropping unit **406**, in one embodiment, is used to crop the captured image. Upon retrieving the captured image from frame buffer **402**, image cropping unit **406** identifies the bounding box containing distorted readout digit value or readout digit image based on the digit location from CPU **416**. After cropping, the cropped image containing distorted readout digit value is forwarded to image rotation unit **408**.

(64) Image rotation unit **406** rotates the distorted readout digit value to a non-distorted or substantially non-distorted readout digit value based on the angle of rotation from CPU **416**. Note that the non-distorted or substantially non-distorted readout digit value should be fairly close to the original readout digit value displayed by the meter. The cropped and rotated readout digit image is forwarded to image scaler **410**.

(65) Image scaler **410** is used to resize or scale the cropped and rotated readout digit image to a predefined size or more desirable scale. After scaling, the scaled readout digit image is forwarded to digit classification model **412**. Note that the scaled readout digit image should be the same or substantially the same as the readout digit value.

(66) Digit classification model **412**, in one aspect, is used to recognize the digits, numbers, and/or dials. Based on the recognition, digit classification outputs digit values one by one and combines into counter value(s) **130** based on the coordinates provided by CPU **416** as indicated by numeral **430**.

(67) In one embodiment, an FPGA is used to host DRM **112**. For example, FPGA can be programmed to perform functions relating to frame buffer **402**, CPU **416**, interfaces **420-422**, image cropping unit **406**, image rotation unit **408**, image scaler **410**, and digit classification model **412**. If wireless communications are used, a wireless FPGA or WAP can be used to host DRM **112**.

(68) An advantage of using DRM **112** is to reduce potential misclassification of readout counter value. For example, numbers 1 and 7 or 6 and 9 are directional sensitive and they can be misclassified due to rotation of digits. As such, cropping, rotating, and scaling can be important for generating accurate counter value **130**.

(69) FIG. 5 is a block diagram illustrating measuring system or AMS **500** using a two-module approach in accordance with one embodiment of the present invention. AMS **500** illustrates a meter **502** with a two-module approach wherein the first module is DRM **508** and the second module is DLM **506**. While DRM **508**, in one embodiment, is permanently coupled to meter **502**, DLM **506** is removed from AMS **500** after calibration. It should be noted that the underlying concept of the exemplary embodiment(s) of the present invention would not change if one or more blocks (ICs or elements) were added to or removed from FIG. 5.

(70) Meter **502** includes a display **516** configured to display or project an original readout counter value **520** which indicates the measured amount of substance or resources such as electricity or water consumed over a period of time. Meter **502** can be an electro-mechanical meter, electricity meter, electric meter, electrical meter, energy meter, kilowatt-hour meter, gas meter, displacement water meter, pressure meter, thermal instrument, heat sensor, or the like. Meter **502** is generally located or installed at the customer's or consumer's premise for metering, measuring, sensing, and/or managing.

(71) Camera **526**, in one embodiment, is placed, fixed, or anchored at a predefined distance from display **516** whereby field of view **118** between display **516** and camera **106** is fixed. Camera **526** can be a semiconductor based camera with lens **528** fabricated in a single chip or IC. Alternatively, camera **526** is constructed as a part of FPGA fabricated on the same die. When field of view **118** includes display **516** showing readout counter value **520**, camera **526** can capture an image based on field of view **118** wherein the image may present a distorted readout counter value or readout counter image **522** depending the distance and/or orientation of lens **528** positioned in front of meter **502**.

(72) DRM **508**, which is a first module, is configured to be permanently installed in the consumer's premise nearby meter **502**. In one embodiment, an FPGA containing configurable logic blocks ("LB") **112** and interface block ("IF") **512** is configured to host or imbed DRM **508**. If DRM **508** requires wireless communication capability, a WAP or wireless FPGA may be used to perform various DRM functions.

(73) DLM **506**, which is a second module, is configured to be removed once the calibration phase is completed. In one embodiment, an FPGA containing configurable logic blocks ("LB") **110** and interface block ("IF") **510** is configured to host DLM **506**. If DLM **506** requires wireless communication capability, a WAP or wireless FPGA may be used to perform various DLM functions. The calibration phase, in one aspect, is a process of identifying coordinates based on the captured image(s) using multi-task model(s). Since multi-task model requires sophisticated circuitry such as ML models and computations, a large FPGA or wireless FPGA may be needed to perform various DLM functions.

(74) In operation, camera **526** captures an image containing readout counter image **522** via field of view **118** focusing on readout counter value **520** projected by display **516**. Readout counter image **522** is subsequently passed to DLM **506** as indicated by numeral **532** via a communication channel **530**. Upon identifying the coordinates and angle of rotation by DLM **506**, the coordinates and angle are transmitted to DRM **508** as indicated by numeral **536**. After retrieving readout counter image **522** from a storage location, DRM **506** restores readout counter image **522** to readout counter value **520** in accordance with the coordinates and angle of rotation. After restoration, the counter value is generated based on the original or similar to original readout counter value.

(75) A benefit of employing AMS **500** is that by attaching DRM to the meter, it modifies a traditional metering instrument into a smart metering instrument capable of meter reading and managing remotely.

(76) An AMS or measuring system **500**, in an alternative embodiment, contains one or more configurable ICs or FPGAs for processing information in accordance with optical images. System **500** includes a metering instrument, a measuring device, and a localization module. The metering instrument such as meter **502** contains at least one display for visually displaying a number and measures the amount of measurable substance passing through the metering instrument within a predefined period. The measuring device such as DRM **508**, including a camera situated in front of the display of the metering instrument, is configured to capture an image of the number representing at least a portion the measurable substance. The localization module such as DLM **506** which is removably coupled to the measuring device is operable to generate the coordinates via an onboard multi-task model block in accordance with the image received from the camera.

(77) In one aspect, the measuring device includes a recognition module such as DRM **508** operable to generate a value in response to the image of the number and coordinates for identifying orientation of the image. The localization module, in one example, includes a camera interface and a controller wherein the camera interface is configured to facilitate removably coupling to the camera via a CL communication for receiving the image from the camera. The controller facilitates to generate the coordinates in accordance with the angle prediction. It should be noted that the recognition module can be configured and embedded in an FPGA.

(78) One embodiment of the presently claimed invention is to use a two-module approach to enhance overall efficiency and simplification. While one module is for digit recognition which is installed on site, the other module is used during installation and calibration. After calibration, the module such as DLM **506**, is removed. DLM **506** employs a multi-task model for detecting the counter and simultaneously predicting the orientation of the counter. The benefit of using the two-module approach is to break down the task into localization and recognition subtasks. Since counter detection is more critical, a more accurate and complicated model in a higher performance computer hardware or chip is more desirable for DLM. A lower performance computer hardware or IC can be used to run the simpler classification models such as DRM for recognizing the digits. For example, DRM is capable of cropping, rotating, and scaling the digit images based on the information provided by DLM whereby DRM is able to accurately recognize every digit and outputs the counter value.

(79) An advantage of employing two-module approach is to simplify onsite installation whereby reducing resources and cost since a large module (i.e., DLM) is removed from AMS after calibration.

(80) FIG. **6** is a block diagram illustrating a wireless accessible PLD (“WAP”) capable of providing wireless FPGA for hosting DLM or DRM in accordance with one embodiment of the present invention in accordance with one embodiment of the present invention. It should be noted that the terms “FPGA” and “PLD” are referring to the same or similar configurable logic device that can be programmed to perform user desirable functions after fabrication. WAP **602** includes wireless communication block (“WCB”) **610** and FPGA **612** which is also known as FPGA die, PLD die, FPGA chip, or PLD chip. FPGA **612** further includes configuration download block (“CDB”) **616** and configurable LBs (logic blocks) **618** wherein configurable LBs **618** can be programmed or configured by bitstream(s), bitstream file(s), or configuration bitstream(s) to perform specific logic function(s). It should be noted that the underlying concept of the exemplary embodiment(s) of the present invention would not change if one or more blocks (circuit or elements) were added to or removed from FIG. **6**.

(81) WCB **610**, in one aspect, is a circuit or component configured to provide or facilitate wireless communication between FPGA **612** and an external remote system. For example, a function of WCB is to provide a wireless communication able to transmit or receive information between WAP **602** and an external remote system via a wireless communication network. The wireless communications network includes, but not limited to, Wi-Fi, Bluetooth, cellular, satellite, and/or radio networks. WCB **610** can be a circuit fabricated within an FPGA. Alternatively, WCB **610** can

be placed in a semiconductor die or chip which can be packaged together with FPGA on a module or package.

(82) FPGA **612** is a semiconductor integrated circuit (“IC”) or die containing various configurable LBs as well as memory cells capable of being programmed by a user after the IC is fabricated. To program or configure an FPGA to perform a user defined logic function, a set of bitstreams may be used to program the FPGA to perform user desirable functions. In one example, FPGA includes arrays of configurable LBs and routing fabric wherein the routing fabric as indicated by numeral **630** further contains various reconfigurable interconnects used to connect various LBs.

(83) WAP **602**, in one example, is a module containing an FPGA **612** and WCB **610** for facilitating wireless bitstream download for programming FPGA as well as debugging FPGA. WCB **610**, in one example, is coupled to CDB **616** via connection **632** for facilitating operation of the programming mode. Similarly, WCB **610** connects to LBs via connection **636** for facilitating operation of the logic access mode. Depending on the applications, connections **632-636** can be merged into one connection. WAP **602**, in another example, includes a wireless module such as WCB **610** and an FPGA die such as FPGA **612** wherein WCB **610** and FPGA **612** are packaged onto a single package.

(84) In an alternative embodiment, an IC is fabricated or manufactured to contain a PSD such as WAP **602** configured to have a wireless block and an FPGA block. WAP **602** includes a set of configurable LBs **618**, routing fabric **630**, CDB **616**, and WCB **610**, wherein configurable LBs **618** are able to be selectively programmed to perform one or more logic functions. Routing fabric **630** is configured to selectively route information between configurable LBs **618** and input/output ports based on a routing configuration bitstream or configuration signals. CDB **616** includes a memory which is used to store configuration data such as bitstream for programming configurable LBs **618** as well as routing fabric **630**.

(85) WCB **610** is operable to facilitate wireless programming or wireless debugging configurable LBs **618** via a wireless communications network. In one example, WCB **610** includes a Wi-Fi transmitter and a Wi-Fi receiver. Alternatively, WCB **610** includes a cellular transmitter and a cellular receiver. WCB **610** can also include a Bluetooth transmitter and a Bluetooth receiver. Depending on the applications, WCB **610** may be configured to select one or more built-in selectable Wi-Fi transceiver and Bluetooth transceiver.

(86) It should be noted that WAP **602** provides an FPGA wireless programming download/debug circuit system containing a wireless module such as WCB **610** and an FPGA module such as LBs **618**. The wireless module, connected with CDB **616**, is used to implement a configuration mode. In one aspect, the wireless module connected to the FPGA logic resource are used to realize the user mode. It should be noted that the two modes can be dynamically switched based on the control signals.

(87) One advantage of using a PSD having a wireless block such as WAP **602** is that it is able to provide a wireless program downloading/debugging a PSD such as an FPGA without a physically connected host system.

(88) FIG. 7 is block diagram illustrating an IC or semiconductor chip containing a WAP **700** capable of facilitating wireless FPGA programming for hosting DLM or DRM in accordance with one embodiment of the present invention in accordance with one embodiment of the present invention. WAP **700** includes WCB **620**, CDB **626**, and configurable LBs **628** wherein configurable LBs **628** are programmed or configured by configuration bitstream(s) to perform configured logic function(s). WAP **606**, in one aspect, is similar to WAP **602** shown in FIG. 6 except that WCB **620**, CDB **626**, and LBs **628** are fabricated on a single IC or chip. It should be noted that the underlying concept of the exemplary embodiment(s) of the present invention would not change if one or more blocks (circuit or elements) were added to or removed from FIG. 7.

(89) Since WCB **620**, CDB **626**, and LBs **628** are placed or fabricated on a single IC, die, and/or chip, the layout of WAP **606** improves circuit integration whereby reducing resource allocation as

well as enhancing efficiency. In one embodiment, WAP **700** is configured to have a programming (or configuration) mode and a user mode (or logic access mode). WCB **620**, in one embodiment, facilitates wireless data transmission between WAP **700** and an external remote system, not shown in FIG. 7. The data transmission includes configuration bitstream for programming/reprogramming FPGA. The data transmission also includes input data from the remote system to WAP **606** or output data from WAP **606** to the remote system.

(90) When the programming mode is activated, WCB **620** downloads netlist or updated configuration bitstream from a remote system via a wireless communications network. The bitstream, also known as bitstream file, is forwarded from WCB **620** to CDB **626** via connection **652**. At least a portion of LBs **628** is subsequently programmed or reconfigured by CDB **626** based on the bitstream. When the logic access mode is active, WAP allows a user or user system to wirelessly access programmed FPGA logic functions. For instance, a user can remotely provide an input to a logic function of FPGA via WCB **620** using connection **656** and/or routing fabric **650**. The result generated by the logic function of FPGA or LBs **628** is subsequently transmitted to the user via WCB **620**. Depending on the applications, connections **650-656** can be merged into one configurable fabric.

(91) One advantage of using WAP **700** is to increase integration and save design costs while improving resource utilization.

(92) FIG. **8** is a diagram illustrating an FPGA containing one or more WCBs capable of being programmed to perform at least some functions of a measuring system in accordance with one embodiment of the present invention. Diagram **800** includes multiple programmable partitioned regions (“PPR”) **802-808**, a programmable interconnection array (“PIA”) **850**, internal power distribution fabric, and regional input/output (“I/O”) ports **866**. PPRs **802-808** further includes control units **810, 820, 830, 840**, memories **812, 822, 832, 842**, configurable WCBs **852-858**, and logic blocks (“LBs”) **816, 826, 836, 846**. Note that control units **810, 820, 830, 840** can be configured into one single control unit, and similarly, memory **812, 822, 832, 843** can also be configured into one single memory device for storing configurations. Furthermore, configurable WCBs **852-858** can also be combined into one single programmable WCB. In one aspect, WAP is a PSD containing at least one FPGA and WCB. It should be noted that the underlying concept of the exemplary embodiment(s) of the present invention would not change if one or more blocks (circuit or elements) were added to or removed from diagram **800**.

(93) LBs **816, 826, 836, 846**, include multiple LABs **818, 828, 838, 848**, wherein each LAB can be further organized to include, among other circuits, a set of programmable logical elements (“LEs”) or macrocells, not shown in FIG. **8**. Each LAB, in one example, may include anywhere from **32** to **512** programmable LEs. I/O pins (not shown in FIG. **8**), LABs, and LEs are linked by PIA **850** and/or other buses, such as buses **862, 814, 824, 834, 844**, for facilitating communication between PIA **850** and PPRs **802-808**.

(94) Each LE includes programmable circuits such as the product-term matrix, and registers. For example, every LE can be independently configured to perform sequential and/or combinatorial logic operation(s). It should be noted that the underlying concept of PSD would not change if one or more blocks and/or circuits were added or removed from PSD.

(95) Control units **810, 820, 830, 840**, also known as configuration logics, can be a single control unit. Control unit **810**, for instance, manages and/or configures individual LE in LAB **818** based on the configuration stored in memory **812**. It should be noted that some I/O ports or I/O pins are configurable so that they can be configured as input pins and/or output pins. Some I/O pins are programmed as bi-directional I/O pins while other I/O pins are programmed as unidirectional I/O pins. The control units such as unit **810** is used to handle and/or manage PSD operations in accordance with system clock signals.

(96) LBs **816, 826, 836, 846** are programmable by the end user(s). Depending on the applications, LBs can be configured to perform user specific functions based on a predefined functional library

facilitated by configuration software. PSD, in some applications, also includes a set fixed circuits for performing specific functions. For example, PSD can include a portion of semiconductor area for a fixed non-programmable processor for enhance computation power.

(97) PIA **850** is coupled to LBs **816**, **826**, **836**, **846** via various internal buses such as buses **814**, **824**, **834**, **844**, **862**. In some embodiments, buses **814**, **824**, **834**, **844**, and **862** are part of PIA **850**. Each bus includes channels or wires for transmitting signals. It should be noted that the terms channel, routing channel, wire, bus, connection, and interconnection are referred to the same or similar connections and will be used interchangeably herein. PIA **850** can also be used to receives and/or transmits data directly or indirectly from/to other devices via I/O pins and LABs.

(98) A function of WCB such as WCB **852** is a special purpose communication unit capable of transmitting or receiving information between WAP and an external remote system via a wireless communication network. A configurable or programmable WCB, in one aspect, is able to be programmed to facilitate wireless communication using a wireless communication, such as, but not limited to, Wi-Fi, Bluetooth, cellular, satellite, and/or radio networks.

(99) An advantage of employing a programmable WCB is to enable AMS to independently communicate with an external system via wireless communication network.

(100) FIG. **9** is a diagram **801** illustrating a routing logic or fabric containing programmable arrays for facilitating interconnecting various components including WCB routing in accordance with one embodiment of the present invention. Diagram **801** includes control logic **8206**, PIA **8202**, I/O pins **8230**, and clock unit **8232**. Control logic **8206**, which may be similar to control units shown in FIG. **8**, provides various control functions including channel assignment, differential I/O standards, and clock management. Control logic **8206** may contain volatile memory, non-volatile memory, and/or a combination of volatile and nonvolatile memory device for storing information such as configuration data. In one embodiment, control logic **8206** is incorporated into PIA **8202**. It should be noted that the underlying concept of the exemplary embodiment(s) of the present invention would not change if one or more blocks (circuit or elements) were added to or removed from diagram **801**.

(101) I/O pins **8230**, connected to PIA **8202** via a bus **8231**, contain multiple programmable I/O pins configured to receive and/or transmit signals to external devices. Each programmable I/O pin, for instance, can be configured to input, output, and/or bi-directional pin. Depending on the applications, I/O pins **8230** may be incorporated into control logic **8206**.

(102) Clock unit **8232**, in one example, connected to PIA **8202** via a bus **8233**, receives various clock signals from other components, such as a clock tree circuit or a global clock oscillator. Clock unit **8232**, in one instance, generates clock signals in response to system clocks as well as reference clocks for implementing I/O communications. Depending on the applications, clock unit **8232**, for example, provides clock signals to PIA **8202** including reference clock(s).

(103) PIA **8202**, in one aspect, is organized into an array scheme including channel groups **8210** and **8220**, bus **8204**, and I/O buses **814**, **824**, **834**, **844**. Channel groups **8210**, **8220** are used to facilitate routing information between LBs based on PIA configurations. Channel groups can also communicate with each other via internal buses or connections such as bus **8204**. Channel group **8210** further includes interconnect array decoders ("IADs") **8212-8218**. Channel group **8220** includes four IADs **8222-8228**. A function of IAD is to provide a configurable routing resources for data transmission.

(104) IAD such as IAD **8212** includes routing multiplexers or selectors for routing signals between I/O pins, feedback outputs, and/or LAB inputs to reach their destinations. For example, an IAD can include up to 36 multiplexers which can be laid out in four banks wherein each bank contains nine rows of multiplexers. It should be noted that the number of IADs within each channel group is a function of the number of LEs within the LAB.

(105) PIA **8202**, in one embodiment, designates a special IAD such as IAD **8218** for handling WCB routing. For example, IAD **8218** is designated to handle connections and/or routings between

WCB and the LABs to facilitate data transmission using a wireless network. It should be noted that additional IADs may be allocated for handling WCB operations.

(106) An advantage of using IAD **8218** within PIA as a designated WCB routing is that it integrates WCB with FPGA to provide network communication using a detected nearby wireless communications network.

(107) The exemplary embodiment of the present invention includes various processing steps, which will be described below. The steps of the embodiment may be embodied in machine or computer-executable instructions. The instructions can be used to cause a general-purpose or special-purpose system, which is programmed with the instructions, to perform the steps of the exemplary embodiment of the present invention. Alternatively, the steps of the exemplary embodiment of the present invention may be performed by specific hardware components that contain hard-wired logic for performing the steps, or by any combination of programmed computer components and custom hardware components.

(108) FIG. **10** is a flowchart **1000** illustrating a process of obtaining a counter value via a measuring system in accordance with one embodiment of the present invention. At block **1002**, a process capable of identifying captured images using configurable ICs is able to capture an image of a sequence of digits displayed by a meter for measuring utility usage via a camera. The process, for example, is capable of activating an onboard camera situated in front of an electricity meter to take a picture containing a sequence of digits representing usage of electricity within a predefined time period.

(109) At block **1004**, the image is forwarded to DLM for calculating coordinates. In one embodiment, a multi-task model is activated to calculate the coordinates for obtaining orientation of the sequence of digits. In one aspect, the process facilitates data transmission via a wireless network through an onboard wireless communication block of an FPGA.

(110) At block **1006**, the image is forwarded to DRM for generating a counter value based on the image and the coordinates. In one example, the process facilitates transmission of the counter value via a wireless network using an onboard WCB of FPGA.

(111) At block **1008**, DLM is subsequently removed from the digit recognition module after the coordinates are computed.

(112) FIG. **11** is a diagram illustrating a digital processing system which can be used as AMS for meter reading employing FPGAs in accordance with one embodiment of the present invention. Computer system **1100** includes a processing unit **1101**, an interface bus **1112**, and an input/output (“IO”) unit **1120**. Processing unit **1101** includes a processor **1102**, main memory **1104**, system bus **1111**, static memory device **1106**, bus control unit **1105**, I/O element **1130**, and FPGA **1185**. It should be noted that the underlying concept of the exemplary embodiment(s) of the present invention would not change if one or more blocks (circuit or elements) were added to or removed from FIG. **11**.

(113) Bus **1111** is used to transmit information between various components and processor **1102** for data processing. Processor **1102** may be any of a wide variety of general-purpose processors, embedded processors, or microprocessors such as ARM® embedded processors, Intel® Core™ Duo, Core™ Quad, Xeon®, Pentium™ microprocessor, Motorola™ 68040, Ryzen™, AMD® family processors, or Power PC™ microprocessor.

(114) Main memory **1104**, which may include multiple levels of cache memories, stores frequently used data and instructions. Main memory **1104** may be RAM (random access memory), MRAM (magnetic RAM), or flash memory. Static memory **1106** may be a ROM (read-only memory), which is coupled to bus **1111**, for storing static information and/or instructions. Bus control unit **1105** is coupled to buses **1111-1112** and controls which component, such as main memory **1104** or processor **1102**, can use the bus. Bus control unit **1105** manages the communications between bus **1111** and bus **1112**. Mass storage memory or SSD which may be a magnetic disk, an optical disk, hard disk drive, floppy disk, CD-ROM, and/or flash memories are used for storing large amounts of

data.

(115) I/O unit **1120**, in one embodiment, includes a display **1121**, keyboard **1122**, cursor control device **1123**, and PLD **1125**. Display device **1121** may be a liquid crystal device, cathode ray tube (“CRT”), touch-screen display, or other suitable display device. Display **1121** projects or displays images of a graphical planning board. Keyboard **1122** may be a conventional alphanumeric input device for communicating information between computer system **1100** and computer operator(s). Another type of user input device is cursor control device **1123**, such as a conventional mouse, touch mouse, trackball, or other type of cursor for communicating information between system **1100** and user(s).

(116) PLD **1125** is coupled to bus **1112** for providing configurable logic functions to local as well as remote computers or servers through wide-area network. PLD **1125** and/or FPGA **1185** includes one or more WCBs for facilitating wireless access. In one example, PLD **1125** may be used in a modem or a network interface device for facilitating communication between computer **1100** and the network. Computer system **1100** may be coupled to a number of servers via a network infrastructure as illustrated in the following discussion.

(117) FIG. **12** is a diagram **1200** illustrating a cloud-based system environment using one or more measuring systems in accordance with one embodiment of the present invention. Diagram **1200** illustrates AI server **1208**, communication network **1202**, switching network **1204**, Internet **1250**, and portable electric devices **1213-1219**. In one aspect, PSD or WAP having various WCBs can be used in AI server, portable electric devices, and/or switching network. Network or cloud network **1202** can be wide area network (“WAN”), metropolitan area network (“MAN”), local area network (“LAN”), satellite/terrestrial network, or a combination of WAN, MAN, and LAN. It should be noted that the underlying concept of the exemplary embodiment(s) of the present invention would not change if one or more blocks (or networks) were added to or removed from diagram **1200**.

(118) Network **1202** includes multiple network nodes, not shown in FIG. **12B**, wherein each node may include mobility management entity (“MME”), radio network controller (“RNC”), serving gateway (“S-GW”), packet data network gateway (“P-GW”), or Home Agent to provide various network functions. Network **1202** is coupled to Internet **1250**, AI server **1208**, base station **1212**, and switching network **1204**. Server **1208**, in one embodiment, includes machine learning computers (“MLC”) **1206**.

(119) Switching network **1204**, which can be referred to as packet core network, includes cell sites **1222-1226** capable of providing radio access communication, such as 3G (3.sup.rd generation), 4G, or 5G cellular networks. Switching network **1204**, in one example, includes IP and/or Multiprotocol Label Switching (“MPLS”) based network capable of operating at a layer of Open Systems Interconnection Basic Reference Model (“OSI model”) for information transfer between clients and network servers. In one embodiment, switching network **1204** is logically coupling multiple users and/or mobiles **1216-1220** across a geographic area via cellular and/or wireless networks. It should be noted that the geographic area may refer to a campus, city, metropolitan area, country, continent, or the like.

(120) Base station **1212**, also known as cell site, node B, or eNodeB, includes a radio tower capable of coupling to various user equipments (“UEs”) and/or electrical user equipments (“EUEs”). The term UEs and EUEs are referring to the similar portable devices and they can be used interchangeably. For example, UEs or PEDs can be cellular phone **1215**, laptop computer **1217**, iPhone® **1216**, tablets and/or iPad® **1219** via wireless communications. Handheld device can also be a smartphone, such as iPhone®, BlackBerry®, Android®, and so on. Base station **1212**, in one example, facilitates network communication between mobile devices such as portable handheld device **1215** or **1219** via wired and/or wireless communications networks. It should be noted that base station **1212** may include additional radio towers as well as other land switching circuitry.

(121) Internet **1250** is a computing network using Transmission Control Protocol/Internet Protocol (“TCP/IP”) to provide linkage between geographically separated devices for communication.

Internet **1250**, in one example, couples to supplier server **1238** and satellite network **1230** via satellite receiver **1232**. Satellite network **1230**, in one example, can provide many functions as wireless communication as well as global positioning system (“GPS”). It should be noted that WAP can be applied to many fields, such as, but not limited to, smartphones **1215-1216**, satellite network **1230**, automobiles **1213**, AI server **1208**, business **1207**, and homes **1220**.

(122) While embodiments of the present invention have been shown and described, it will be obvious to those of ordinary skills in the art that based upon the teachings herein, changes and modifications may be made without departing from this exemplary embodiment(s) of the present invention and its broader aspects. Therefore, the appended claims are intended to encompass within their scope all such changes and modifications as are within the true spirit and scope of this exemplary embodiment(s) of the present invention.

Claims

1. A method for capturing and adjusting an image via a configurable integrated circuit (“IC”) chip, comprising: capturing an image representing a measurement displayed by a measuring meter via a camera mounted within vicinity of the measuring meter; retrieving coordinates and angle prediction from a local memory in accordance with the image; cropping boundaries of the image based on the coordinates and rotating the image based on the angle prediction to generate an adjusted image; transmitting a counter value in response to the adjusted image via a wireless network; supplying the coordinates from the localization module to the recognition module; generating coordinates at least partially based the angle prediction; forwarding the image to a digit recognition module for generating the counter value based on the image and the coordinates; and removing a digit localization module from a digit recognition module after the coordinates are computed.
2. The method of claim 1, further comprising capturing a sequence of digits representing utility usage by an image capturing device via a wireless communication network.
3. The method of claim 1, further comprising forwarding the image to a digit localization module for calculating coordinates.
4. The method of claim 1, further comprising detecting at least one digit from the image and an angle prediction.
5. The method of claim 1, wherein capturing an image includes activating an onboard camera situated in front of an electricity meter to take a picture containing a sequence of digits representing usage of electricity within a predefined time period.
6. The method of claim 1, further comprising forwarding the image to a digit localization module and activating a multi-task model to calculating coordinates for obtaining orientation of a sequence of digits.
7. The method of claim 1, further comprising forwarding the image to a digit localization module and facilitating data transmission via a wireless network through an onboard wireless communication block of a field programmable gate array (“FPGA”).
8. The method of claim 1, further comprising forwarding the image to a digit recognition module and facilitating transmission of the counter value via the wireless network through an onboard wireless communication block of a field programmable gate array (“FPGA”).
9. The method of claim 1, further comprising transmitting the coordinates from a digit localization module to a digit recognition module for generation of the counter value.
10. A system containing configurable integrated circuits (“IC”) for processing information via captured images, the system comprising: a metering instrument containing at least one display for visually displaying a number representing amount of measurable substance within a predefined time period; a camera coupled to the display of the metering instrument and configured to capture an image of the number representing at least a portion of the amount of measurable substance; a localization module removably coupled to the camera and operable to generate the coordinates in

accordance with the image received from the camera, the localization module including, a multi-task model block operable to detect at least one digit from the image and an angle prediction, a controller configured to generate a set of coordinates at least partially based the angle prediction; and a digit recognition module coupled to the localization module and configured to generating a counter value based on the image and the set of coordinates.

11. The system of claim 10, further comprising a recognition module coupled to the camera and operable to generate a counter value in response to the image and coordinates utilized for minimizing image distortion.

12. The system of claim 10, wherein the localization module further includes a recognition interface configured to facilitate removably coupling to a localization-to-recognition communication for supplying the coordinates from the localization module to the recognition module.

13. The system of claim 10, wherein the localization module further includes, an image cropping block operable to crop boundaries of the image based on the coordinates, and an image rotation block operable to rotate the image based on the angle prediction.

14. The system of claim 10, further comprising a field-programmable gate array ("FPGA") configured to be programmed at least a portion of the FPGA to perform functions of the localization module.

15. An apparatus of capturing and adjusting an image via a configurable integrated circuit ("IC") chip, comprising: means for capturing an image representing a measurement displayed by a measuring meter via a camera mounted within vicinity of the measuring meter; means for retrieving coordinates and angle prediction from a local memory in accordance with the image; means for cropping boundaries of the image based on the coordinates and rotating the image based on the angle prediction to generate an adjusted image; means for transmitting a counter value in response to the adjusted image via a wireless network; means for supplying the coordinates from the localization module to the recognition module; means for generating coordinates at least partially based the angle prediction; means for forwarding the image to a digit recognition module for generating the counter value based on the image and the coordinates; and means for removing a digit localization module from a digit recognition module after the coordinates are computed.

16. The apparatus of claim 15, further comprising means for capturing a sequence of digits representing utility usage by an image capturing device via a wireless communication network.

17. The apparatus of claim 15, further comprising means for forwarding the image to a digit localization module for calculating coordinates.

18. The apparatus of claim 15, further comprising means for detecting at least one digit from the image and an angle prediction.

19. The apparatus of claim 15, wherein means for capturing an image includes means for activating an onboard camera situated in front of an electricity meter to take a picture containing a sequence of digits representing usage of electricity within a predefined time period.

20. The apparatus of claim 15, further comprising means for forwarding the image to a digit localization module and activating a multi-task model to calculating coordinates for obtaining orientation of a sequence of digits.
